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Symptoms based Disease prediction Web Application

ABSTRACT

The way people seek health-related information is undergoing significant changes worldwide. One major challenge is the difficulty individuals face in finding trustworthy online resources about illnesses, diagnoses, and treatment options. A solution lies in creating a recommendation system for doctors and medications that utilizes review mining to simplify this process and save time. Four machine learning algorithms are used together to achieve around 96% accuracy. It provides departments and doctors information based on prediction to users. It focuses on securely storing patient history for future references integrating advanced password hashing techniques as well as JWT (Jason Web Token) based authentication. It actually aims to improve healthcare accessibility to reduce patient confusion for faster as well as informed decision making. However, understanding complex medical language can be overwhelming for non-specialist users. The vast amount of medical data is there across different platforms which further complicates their search. To address these issues, such a system must be designed to cater specifically to the needs of the healthcare sector, ensuring it remains user-friendly and easy to understand for all individuals.

Keywords: Health sector, Disease's diagnosis, securing patient history, Symptoms based diseases prediction.

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Chapter – 1: INTRODUCTION

1.1 BACKGROUND

Digital technology now influences nearly every industry and healthcare is certainly not the exception. Individuals are turning to the internet as their primary source of information concerning their health, its symptoms, and available treatments. The expansion of internet services, coupled with the increasing use of smartphones, grants users instant access to a plethora of health-related information. Nevertheless, there is still an incredible discrepancy between the information that can be found, and the understanding and trust a user can place in that information.

The most innovative approaches to AI and ML are being presented in the medical domain, from helping identify illnesses to helping create medications. Simple enough for a non-professional to understand, healthcare information is needed. AI and patient-centred, automated services such as therapy bots or healthcare apps that utilize machine learning could provide the solutions for better information comprehension.

1.2 PROBLEM STATEMENT

Nowadays, everyone is turning to the internet to seek information regarding diseases, diagnoses, and treatments. They aim to familiarize themselves with relevant healthcare topics. However, the large amounts of data spread across the internet are difficult to manage, as the information is both vast and advanced. With little medical knowledge, laypeople often struggle with the diagnosis themselves because of the complicated medical jargons. To make things worse, there is no single comprehensive guide or repository which puts everything together in an orderly manner, making navigation difficult and wasting one's time in understanding the basics first.

There is no recommended system that provides users with personalized recommendations for doctors and medications through review mining techniques. Such a system should not only be able to assist with the understanding of complex medical information, but also integrate with other features of the healthcare system, while being user-friendly for all types of users regardless of their medical education level. Solving these interface design problems will make accessing health information more effective and accurate and assist users in navigation and retrieval of health information.

1.3 AIMS AND OBJECTIVES

This project's goal is to design an interactive intelligent secure web application that can predict potential diseases from a given user symptom list. The platform will provide guide medical assistance aid to users empowering them medically navigate their health care with preliminary guidance.

The goals include the following:

- Gather a reliable dataset containing diseases and their corresponding symptoms, then prepare the data for model training.
- Applying different model algorithms such as Decision Tree, Random Forest, K-Nearest Neighbours (KNN), and Gaussian Naive Bayes to perform disease predictions and evaluate their prediction accuracies.
- Develop a secure, responsive, and a user-friendly web application that allows for easy input of symptoms and provides automatically controlled data access.
- Create a systematic user management system that allows for easy storing and accessing prior user prediction records which is useful for potential future research.
- Mention appropriate medical specialists or doctors to users for timely professional medical intervention based on predicted diseases.
- Combine outputs of multiple models to improve prediction accuracy and reduce errors.

With these goals, it's possible to provide a system that aids in rapid access to healthcare services while giving automated disease prediction assistance.

1.4 THESIS LAYOUT

This report has been structured in a way that captures the stages of research, design and development that were carried out throughout the entire project.

Chapter 1: An introduction to the topic is provided in which emphasizes the importance of a cognitive-friendly disease forecasting tool.

Chapter 2: the literature containing information is explained, as well as the gaps in research and previously done works pertinent to the fusion of healthcare and machine learning.

Chapter 3: The approach used on data collection, model development, and system decomposition are discussed in the methodology details in that section.

Chapter 4: It consists of the various experimental results, analysis of model performance, and system evaluations.

Chapter 5: Some key learnings and challenges faced with concluding remarks are discussed in that section.

Chapter 6: the entire body of work done is encapsulated, along with the instruction for further development and additional research.

Chapter – 2: LITERATURE REVIEW

2.1 HEALTH INFORMATION ACCESSIBILITY CHALLENGES

While the internet has a lot of health-related content, users struggle evaluating trustworthiness and interpreting complicated medical languages. More often than not, people with no prior training misread medical jargons and oversimplify complex symptoms. Many self-diagnose and their inaccurate interpretation can be potentially hazardous. Most online health systems bombard their users with ambiguous, non-verified data. Health information needs validation, simplification and organization into digestible formats. Our project seeks to fill the existing void in platforms that accurately diagnose medical issues based on reported bodily symptoms and efficiently suggest professional medical care.

2.2 MACHINE LEARNING IN HEALTHCARE

One of the facets of modern healthcare is machine learning, which offers sophisticated solutions for medical diagnosis, designing prognosis and treatment plans [2]. Decision Trees lays down the routes of the decision making, while Random Forests increases accuracy through ensemble learning. KNN thrives with new data, and Naive Bayes offers speedy and simple classification based on the assumed probability. Application of these models in healthcare is centred on the ability to predict diseases formed from patterns of symptoms provided by patients.

2.3 EXISTING SYSTEMS AND RESEARCH GAPS

The symptom checking services provided by platforms such as WebMD and Ada Health propose possible conditions, but do not provide personalization or recommend doctors. In addition, many do not justify how the forecasts are made eliminating trust from the users. Most research concentrates on single diseases rather than predicting multiple diseases [3] from a symptoms analysis. These gaps are closed with our proposed system that provides personalized prediction structure, specialized guidelines, and a protected design which puts the user first.

Chapter – 3: METHODOLOGY

3.1 DATA COLLECTION AND PREPROCESSING

The data I worked with that was sourced from the Columbia University Disease-Symptom Knowledge Base dataset [1], which stems from research done at the New York Presbyterian Hospital. This dataset contains 4920 records and 133 attributes, where each row corresponds to a case of a patient and each column represents a symptom or a prognosis. During pre-processing, we noticed that some symptoms did not contribute meaningfully to making a diagnosis. Therefore, we carried out feature selection by focusing on symptoms that impacted predicting the prognosis. In this case, we managed to reduce the number of features related to symptoms from 133 to 94 relevant symptoms while increasing the model's speed and accuracy.

In addition, there were 41 unique diseases in the prognosis column. To aid in simplifying multi-class classification, they were all mapped to a unique integer identifier from zero to forty. Some of the basic pre-processing tasks included:

- Dealing with missing values by discarding records that were not complete.
- Standardization of categorical values so that models would be trained properly.
- Encoding diseases name that means converting them into numerical values.
- Dividing the entire dataset into training and testing datasets (usually 80% and 20%, respectively) so that it would help to validate the quality of training.

This feature-engineered dataset is what I used to train the machine learning models, ensuring that all relevant information was preserved during the pre-processing stage so the models trained effectively without extraneous data interference.

3.2 MACHINE LEARNING MODELS AND TRAINING

The algorithms we utilized to predict diseases through listed symptoms include: Decision Tree, Random Forest, K-Nearest Neighbours (KNN), and Gaussian Naive Bayes. We implemented and trained them all. Each model has its own strengths, and their combination ensures higher reliability.

Decision Tree Classifier: Decision Trees are flowchart-like structures where symptoms are internal nodes, decision rules are branches, and every leaf node represents a disease. It is simple to interpret and train rapidly, but it can be overfit on noisy data. Moreover, overfitting is common with decision trees when utilizing noisy data.

Random Forest Classifier: This approach is an ensemble of multiple Decision Trees where each tree is trained on features and data samples at random. To get the final prediction or probability, voting procedure is used. Random forest classifiers decrease overfitting while improving generalization consistently.

K-Nearest Neighbours (KNN): This model is non-parametric. A sample is classified to a respective class according to lowest distance among all calculated distances with all classes. Generally, KNN helps in classification or disease prediction if close-symptom patterns cluster in the feature space.

Gaussian Naive Bayes: An example of a Bayes' theorem that assumes independence between features can be Gaussian naive Bayes. Irrespective of this strong assumption, marriage to healthcare classification yields favourable outcomes for the model, thanks to the probabilistic relationship between diseases and symptoms.

15 Accuracy, precision, recall and f1-score metrics are calculated for all models as well as cross-validation techniques are implemented to ensure robustness. The accuracy was estimated around 96% which ensures that all models are suitable for deploying in application.

3.3 WEB APPLICATION ARCHITECHTURE

The web application was created with user interaction and security in mind. It is designed by combining front end and back end together [4]. On the other hand, front end and back end are developed separately.

Frontend:

- It is implemented with React.js a JavaScript based framework where HTML5, CSS3, and JavaScript components are used.
- Secure login and registration pages are designed.

- Symptom form is designed with five selectable symptoms.
- According to prediction result, corresponding departments' name and associated doctors' name are shown in a suitable manner.

Backend:

- Python programming language-based framework, Django is used to design backend.
- In the backend, user authentication is implemented through secure session management via JWT (JSON Web Tokens).
- Handles symptom inputs and queries machine learning models for predictions.
- Stores prediction history in relational databases such as PostgreSQL or MySQL.
- Sperate APIs for registration, login, password reset, and disease prediction are designed.

3.4 WORKFLOW DIAGRAM

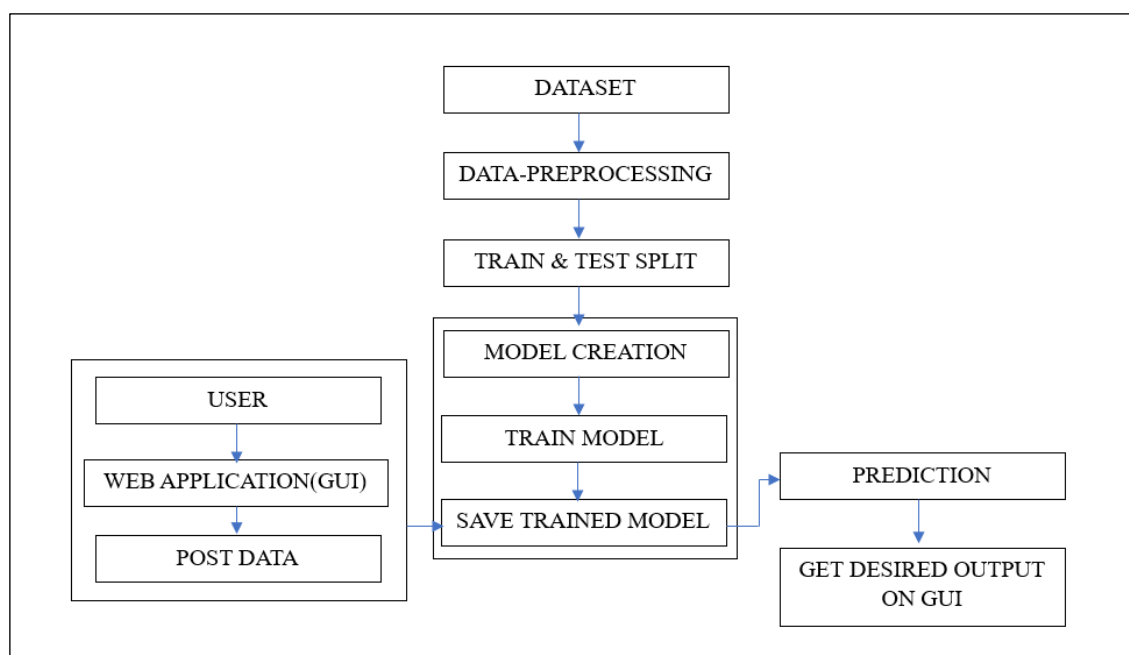


Figure 1: Work flow diagram 1st figure in chapter 3

Chapter – 4: RESULT AND ANALYSIS

Dataset for this project was collected from a study of university of Columbia performed at New York Presbyterian Hospital during 2004. Link of dataset is given below.

<http://people.dbmi.columbia.edu/~friedma/Projects/DiseaseSymptomKB/index.html>

4.1 Model part:

```
In [2]: df=pd.read_csv(r'C:\Users\soham\OneDrive\Desktop\All files\Diseases prediction datasets\training.csv')

In [175]: df.shape

Out[175]: (4920, 133)

In [3]: df.head()

Out[3]:
```

	blackheads	scurring	skin_peeling	silver_like_dusting	small_dents_in_nails	inflammatory_nails	blister	red_sore_around_nose	yellow_crust_ooze	prognosis
...	0	0	0	0	0	0	0	0	0	Fungal infection
...	0	0	0	0	0	0	0	0	0	Fungal infection
...	0	0	0	0	0	0	0	0	0	Fungal infection
...	0	0	0	0	0	0	0	0	0	Fungal infection
...	0	0	0	0	0	0	0	0	0	Fungal infection

Figure 2: 1st figure in chapter 4

The whole train dataset has 4920 numbers of records with 133 numbers of columns which represent symptoms. Some symptoms are not giving any impact on the prognosis column which is our target column. So, I need to extract only those columns or symptoms which is impacting the prognosis.

```
In [202]: print(symptoms_list)

['back_pain', 'constipation', 'abdominal_pain', 'diarrhoea', 'mild_fever', 'yellow_urine', 'yellowing_of_eyes', 'acute_liver_failure', 'swelling_of_stomach', 'swelled_lymph_nodes', 'malaise', 'blurred_and_distorted_vision', 'phlegm', 'throat_irritation', 'redness_of_eyes', 'sinus_pressure', 'runny_nose', 'congestion', 'chest_pain', 'weakness_in_limbs', 'fast_heart_rate', 'pain_during_bowel_movements', 'pain_in_anal_region', 'bloody_stool', 'irritation_in_anus', 'neck_pain', 'dizziness', 'cramps', 'bruising', 'obesity', 'swollen_legs', 'swollen_blood_vessels', 'puffy_face_and_eyes', 'enlarged_thyroid', 'brittle_nails', 'swollen_extremeties', 'excessive_hunger', 'extra_marital_contacts', 'drying_and_tingling_lips', 'slurred_speech', 'knee_pain', 'hip_joint_pain', 'muscle_weakness', 'stiff_neck', 'swelling_joints', 'movement_stiffness', 'spinning_movements', 'loss_of_balance', 'unsteadiness', 'weakness_of_one_body_side', 'loss_of_smell', 'bladder_discomfort', 'foul_smell_of_urine', 'continuous_feel_of_urine', 'passage_of_gases', 'internal_itching', 'toxic_look(typhos)', 'depression', 'irritability', 'muscle_pain', 'altered_sensation', 'red_spots_over_body', 'belly_pain', 'abnormal_menstruation', 'dischromic_patches', 'watering_from_eyes', 'increased_appetite', 'polyuria', 'family_history', 'mucoid_sputum', 'rusty_sputum', 'lack_of_concentration', 'visual_disturbances', 'receiving_blood_transfusion', 'receiving_unsterile_injections', 'coma', 'stomach_bleeding', 'distention_of_abdomen', 'history_of_alcohol_consumption', 'fluid_overload', 'blood_in_sputum', 'prominent_veins_on_calf', 'palpitations', 'painful_walking', 'pus_filled_pimples', 'blackheads', 'scurring', 'skin_peeling', 'silver_like_dusting', 'small_dents_in_nails', 'inflammatory_nails', 'blister', 'red_sore_around_nose', 'yellow_crust_ooze']

In [203]: print(len(symptoms_list))

94
```

Figure 3: 2nd figure in chapter 4

This is the extracted symptoms list with 94 symptoms. I have used that list of symptoms to train models.

```
In [228]: print(disease)

['Fungal infection', 'Allergy', 'GERD', 'Chronic cholestasis', 'Drug Reaction', 'Peptic ulcer disease', 'AIDS', 'Diabetes ', 'Gastroenteritis', 'Bronchial Asthma', 'Hypertension ', 'Migraine', 'Cervical spondylosis', 'Paralysis (brain hemorrhage)', 'Jaundice', 'Malaria', 'Chicken pox', 'Dengue', 'Typhoid', 'hepatitis A', 'Hepatitis B', 'Hepatitis C', 'Hepatitis D', 'Hepatitis E', 'Alcoholic hepatitis', 'Tuberculosis', 'Common Cold', 'Pneumonia', 'Dimorphic hemmorhoids(piles)', 'Heart attack', 'Varicose veins', 'Hypothyroidism', 'Hyperthyroidism', 'Hypoglycemia', 'Osteoarthritis', 'Arthritis', '(vertigo) Paroymsal Positional Vertigo', 'Acne', 'Urinary tract infection', 'Psoriasis', 'Impetigo']

In [229]: print(len(disease))

41
```

```
In [5]: df.replace({'prognosis':{'Fungal infection':0,'Allergy':1,'GERD':2,'Chronic cholestasis':3,'Drug Reaction':4,'Peptic ulcer disease':5,'AIDS':6,'Diabetes ':7,'Gastroenteritis':8,'Bronchial Asthma':9,'Hypertension ':10,'Migraine':11,'Cervical spondylosis':12,'Paralysis (brain hemorrhage)':13,'Jaundice':14,'Malaria':15,'Chicken pox':16,'Dengue':17,'Typhoid':18,'hepatitis A':19,'Hepatitis B':20,'Hepatitis C':21,'Hepatitis D':22,'Hepatitis E':23,'Alcoholic hepatitis':24,'Tuberculosis':25,'Common Cold':26,'Pneumonia':27,'Dimorphic hemmorhoids(piles)':28,'Heart attack':29,'Varicose veins':30,'Hypothyroidism':31,'Hyperthyroidism':32,'Hypoglycemia':33,'Osteoarthritis':34,'Arthritis':35,'(vertigo) Paroymsal Positional Vertigo':36,'Acne':37,'Urinary tract infection':38,'Psoriasis':39,'Impetigo':40}},inplace=True)
df.head()
```

...	blackheads	scurring	skin_peeling	silver_like_dusting	small_dents_in_nails	inflammatory_nails	blister	red_sore_around_nose	yellow_crust_ooze	prognosis
...	0	0	0	0	0	0	0	0	0	0
...	0	0	0	0	0	0	0	0	0	0
...	0	0	0	0	0	0	0	0	0	0
...	0	0	0	0	0	0	0	0	0	0
...	0	0	0	0	0	0	0	0	0	0

Figure 4: 3rd figure in chapter 4

There is total 41 diseases in the prognosis column. It is a classification problem. So, I have assigned numbers from 0 to 40 to all individual diseases. After that I have implemented same pre-processing steps for the test dataset and trained four models which are **Decision tree model**, **Random Forest model**, **Knn model** and **Gaussian naïve bayes model**.

Decision Tree Algorithm

```
In [223]: # from sklearn import tree
# from sklearn.metrics import classification_report, confusion_matrix, accuracy_score, precision_score, recall_score, f1_score

clf3 = tree.DecisionTreeClassifier()
clf3 = clf3.fit(X,y)

y_pred=clf3.predict(X_test)
print("Decision Tree")
print("Accuracy")
print(accuracy_score(y_test, y_pred))
print("precision_score")
print(precision_score(y_test, y_pred, average='macro'))
print("recall_score")
print(recall_score(y_test, y_pred, average='macro'))
print("f1_score")
print(f1_score(y_test, y_pred, average='macro'))

print("Confusion matrix")
conf_matrix=confusion_matrix(y_test,y_pred)
print(conf_matrix)

symptoms = ['abdominal_pain', 'diarrhoea', 'mild_fever', 'yellow_urine', 'cramps']
for k in range(0,len(symptoms_list)):
    for j in symptoms:
        if(z==symptoms_list[k]):
            l2[k]=1

inputtest = [12]
predict = clf3.predict(inputtest)
predicted=predict[0]
result[0]=predicted

h='no'
for a in range(0,len(disease)):
    if(predicted == a):
        h='yes'
        break

if (h=='yes'):
    print(disease[a])

Decision Tree
Accuracy
0.9512195121951219
precision_score
0.926829268292683
recall_score
0.9512195121951219
f1_score
0.9349593495934958
Confusion matrix
[[1 0 ... 0 0 0]
 [0 1 0 ... 0 0 0]
 [0 0 1 ... 0 0 0]
 ...
 [0 0 0 ... 1 0 0]
 [0 0 0 ... 0 1 0]
 [0 0 0 ... 0 0 1]]
Gastroenteritis
```

Figure 5: 4th figure in chapter 4

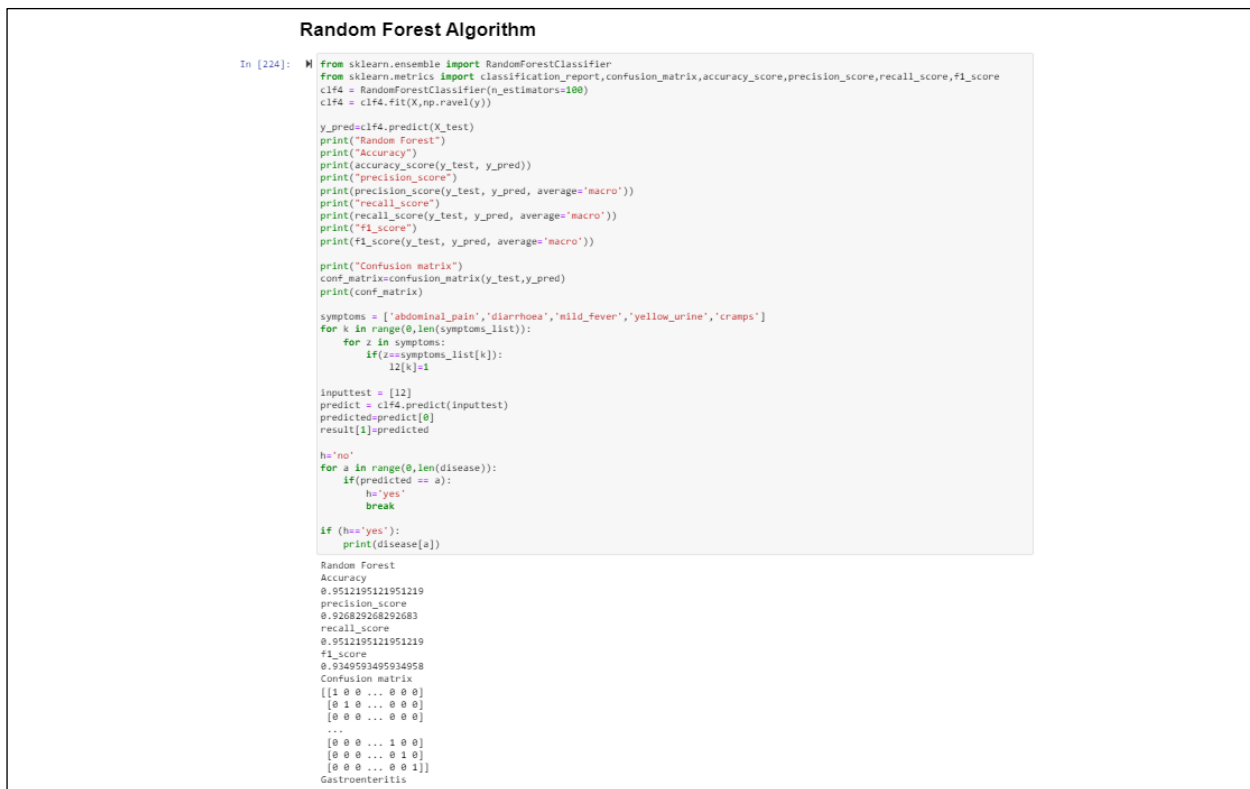


Figure 6: 5th figure in chapter 4

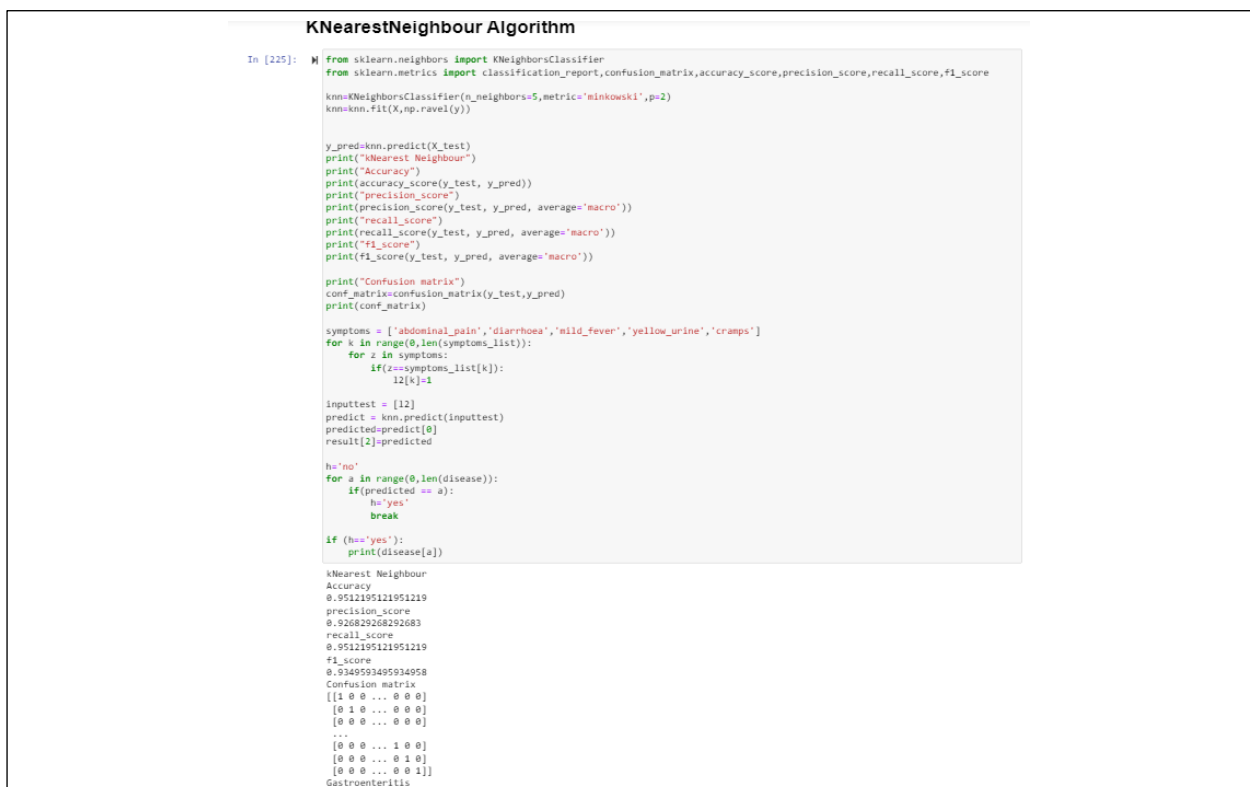


Figure 7: 6th figure in chapter 4

Naive Bayes Algorithm

```
In [226]: from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score, precision_score, recall_score, f1_score

gnb = GaussianNB()
gnb.fit(X, np.ravel(y))

y_pred = gnb.predict(X_test)
print("Naive Bayes")
print("Accuracy")
print(accuracy_score(y_test, y_pred))
print(precision_score(y_test, y_pred, average='macro'))
print(recall_score(y_test, y_pred, average='macro'))
print(f1_score(y_test, y_pred, average='macro'))

print("Confusion matrix")
conf_matrix = confusion_matrix(y_test, y_pred)
print(conf_matrix)

symptoms = ['abdominal_pain', 'diarrhoea', 'mild_fever', 'yellow_urine', 'cramps']

for k in range(0, len(symptoms_list)):
    for z in symptoms:
        if (z == symptoms_list[k]):
            12[k]=1

inputtest = [12]
predict = gnb.predict(inputtest)
predicted=predict[0]
result[3]=predicted

h='no'
for a in range(0, len(disease)):
    if (predicted == a):
        h='yes'
        break
if (h=='yes'):
    print(disease[a])

Naive Bayes
Accuracy
0.9512195121951219
0.926829268292683
recall_score
0.9512195121951219
f1_score
0.9349593495934958
Confusion matrix
[[1 0 0 ... 0 0 0]
 [0 1 0 ... 0 0 0]
 [0 0 1 ... 0 0 0]
 ...
 [0 0 0 ... 1 0 0]
 [0 0 0 ... 0 1 0]
 [0 0 0 ... 0 0 1]]
Typhoid
```

```
In [227]: unique_elements = list(set(result))
print("You have been diagnosed with")
j=1
for i in unique_elements:
    print(f"{j}. {disease[i]}")
    j += 1
result = [0, 0, 0, 0]
```

You have been diagnosed with
1. Gastroenteritis
2. Typhoid

Figure 8: 7th figure in chapter 4

We can see that every model gives accuracy score around 95%, precision score around 92%, recall score around 95% and f1 score around 93%.

Here I am using four models to avoid any kind of miss prediction when patient is getting information regarding departments and doctors to whom he or she should contact. Usually, any patient fill symptoms as per his or her feeling that sometimes miss interprets the prediction of the diseases. In real life scenario, predictions of four models will cover all cases which can lead to miss prediction.

That's how patient can have maximum four departments' doctors' choices in the worst case. On average, patient will get one or two departments' doctors' information. That can speed up the treatment process.

4.2 Web Application part:

4.2.1 Frontend:

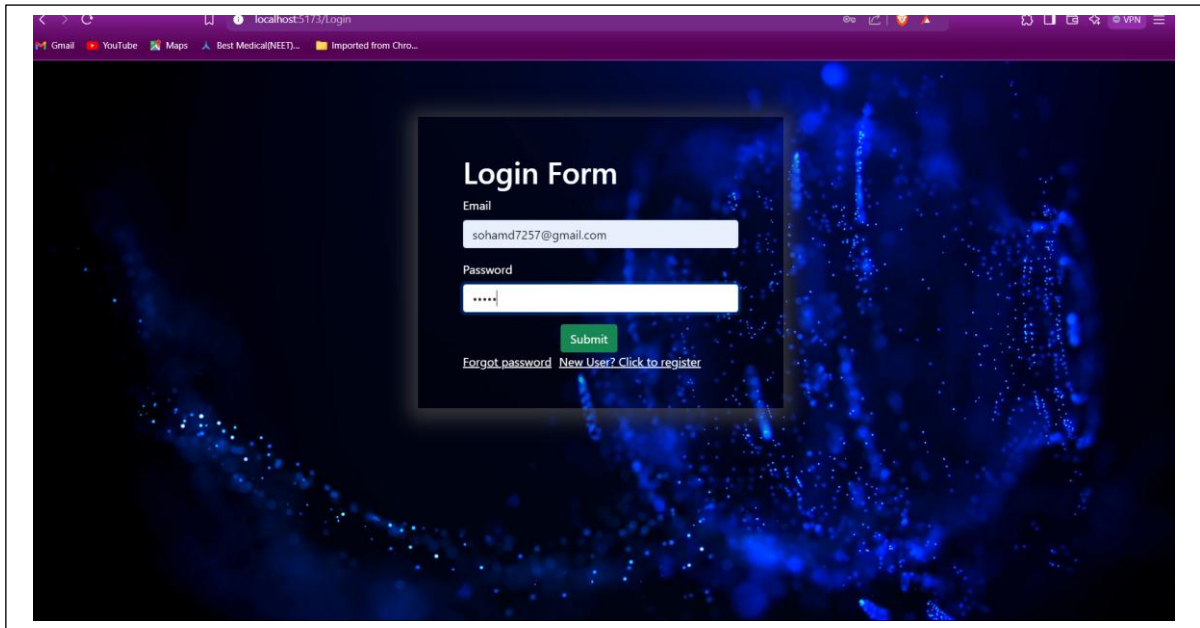


Figure 9: 8th figure in chapter 4

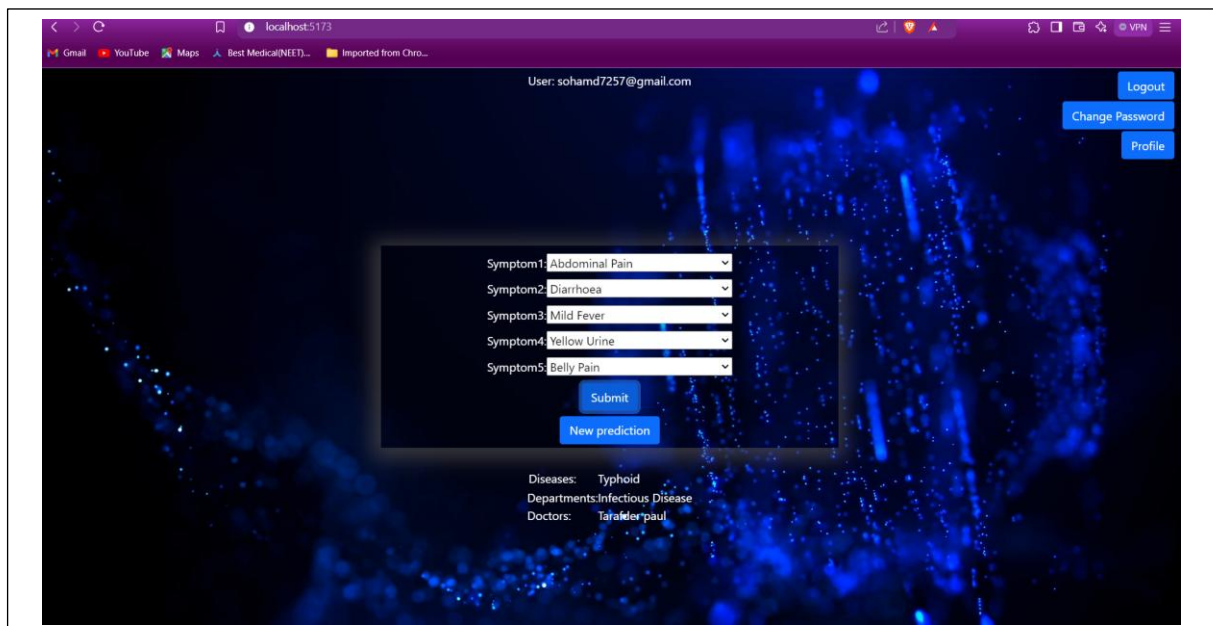


Figure 10: 9th figure in chapter 4

There are Login page, Register page, Prediction page. Patient can login or register using email id & password. After login, patient can fill the form where five symptoms have to be filled to get the prediction.

Here in the example, patient is getting the name of the predicted diseases with departments and doctors' information but in real scenario, patient will only get the information regarding

departments and doctors. Predicted history will be saved at database with patient details, all filled symptoms, departments and doctors' names.

4.2.2 Backend:

On right side, the folder structure of the backend is given.

On the backend, I made two applications to manage all APIs regarding user account management and diseases prediction.

I have use JWT (JSON Web Token) to manage sessions, reset-password both for after and before login. **User gets password reset-link to registered email id.** All passwords of individual user are saved in the database in a hashed form. I have created proctored routes which allow user to surf pages as a logged in user. Without login, user can only access login & register pages.

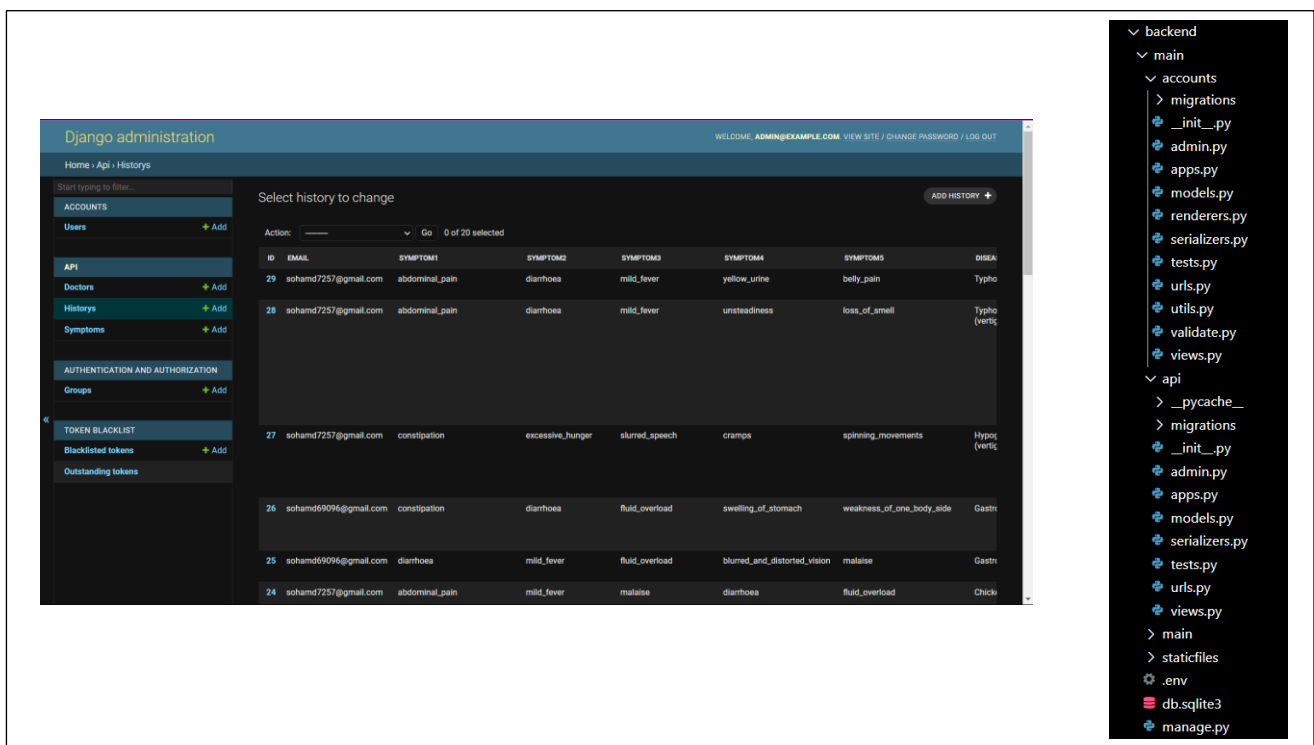


Figure 11: 10th figure in chapter 4

In the Django admin panel, admin can access all history created by all patients. Those histories can be used for future studies.

Chapter – 5: DISCUSSION AND CONCLUSION

5.1 DISCUSSION

The implementation of the Symptoms Based Disease Prediction Web Application showcases the potential impacts that machine learning can have in the real world, especially in the healthcare industry. One milestone of the work was constructing a multi-model architecture aimed at predicting diseases with high accuracy based on symptoms provided by users. The performance metrics of the four models published confirm the accuracy and efficiency machine learning offers for predictive endeavours. In any case, although the technical performance was provided a lot of hope, the greatest takeaway from this project, at least for me, is how it can be applied in the real world. Patients have a hard time of determining which symptoms warrant attention and which doctor or unit they want to see. Our system fills this gap by providing a diagnostic system which will provide department suggestion that stands to significantly reduce the time taken to receive critical care. Some of the salient points learned from the entire endeavours are as follows:

- Quality of data matters. The fact that cleaning the data set and choosing relevant features improves models is enough to prove data quality.
- Using more than one model is useful. It creates redundancy, which, on a broader scope, increases accuracy. In this case, there is decreased instance of incorrect prediction that is crucial in health care.
- User experience is just as important as accuracy. We focused on providing sane interfaces and functional logins while keeping usability in mind for non-technical users.

On the other hand, I do accept the following limitations:

- The dataset, while vetted by medical professionals, is not comprehensive. There are conditions not covered in the dataset which cannot be predicted.
- The system does not take into account the intensity or length of symptoms- which are crucial for diagnosis.
- There is no live doctor or real-time consultation available; guidance is only provided toward relevant departments.

In spite of these difficulties, the application achieves its primary goal of helping the user make the initial move towards sought after medical attention by providing- through artificial intelligence- insightful assistance.

5.2 CONCLUSION

In conclusion, this project is designed, developed, and deployed as a functioning web-based system that can predict probable diseases using multiple machine learning models and allows the user to enter their symptoms. The results obtained were predictive and justified the approach taken during development. The web application enables users to interact with the system and, easily input symptoms. Machine learning model provides predictions on what diseases the user may have given. Recommendations are on which departments and specialists the user should visit.

Moreover, the system ensures secure storage of the prediction history, and authentication is protected using JWT tokens, and passwords are maintained in hashed form for privacy protecting data confidentiality and integrity. The application does not attempt to replace medical diagnosis; however, it tries to facilitate in aiding the user seek professional consultation as a preliminary advisory tool. Incorporation into actual healthcare systems – such as a Telemedicine interface or a hospital triage system – could serve as a convincing next step.



Chapter – 6: SUMMARY AND FUTURE WORK

6.1 SUMMARY

The goal of this project was to create a web-based diseases prediction system aimed at helping users interpret possible health issues from self-reported symptoms. The work started with an exploration of a gap in healthcare: the usability and availability of disease screening tools that are convenient, reliable, and simple to navigate. A quality dataset obtained from a study at University of Columbia, was cleaned and prepared for the application of machine learning. Out of 133 presenting symptoms, 94 pertinent features were selected which left a dataset that was adequately precise for model training. Subsequently, four machine learning algorithms were applied – Decision Tree, Random Forest, K-Nearest Neighbours (KNN), and Gaussian Naive Bayes – then it was verified using set evaluation criteria.

All models recorded an average accuracy of above 95%, which was highest in Random Forest. The models executed were proved to be more dependable because they performed precisely under circumstances whereby symptoms were common to multiple conditions. In order to provide users with an enhanced, yet secure experience, a web application was developed having Django as the backend, while HTML, CSS, and JavaScript were used on the frontend. The system allows people to register and provide their symptoms, obtain predictions, and receive directions to the appropriate department or specialist. All data provided is safely stored with password hashing and session management using JWT.

6.2 FUTURE WORK

In this work, still there are many scopes where improvement work can be done to deliver more valuable service towards end users.

- With a large dataset with more diverse and correlated information, more complex deep learning model can be trained to achieve higher prediction accuracy for complex and rare conditions [5].
- A feature can be implemented, which will suggest next symptoms to the user according to the already selected symptoms. That will surely enhance user experience as well as will increase overall prediction accuracy.
- With the integration of IOT devices, real time health monitoring feature can be enabled which will definitely increase overall prediction accuracy.

Chapter – 7: REFERENCES

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